Mohsin Ali Mohammed

San Diego, CA | +1 (619) 273-7627 | mohammedmohsin5431@gmail.com
LinkedIn | GitHub | Google Scholar | quantdecisionlabs.com | Portfolio

Applied AI engineer and researcher with **4+ years building production LLM, agentic, and ML systems**, and founder of **Quant Decision Labs**, an AI market-reasoning platform combining multi-agent LLM debate, a deterministic risk engine, and a **risk-normalized (R-multiple) evaluation framework** running against live market data in production. Published at **AAAI, EMNLP, and HICSS**. Builds AI systems that are **measured, audited, and falsifiable**, not just shipped.

EXPERIENCE

Quant Decision Labs (Founder) — **2025 – Present**
AI Quant Research Platform: Numin (quantdecisionlabs.com · numin.trade) — *San Diego, CA*

- Designed and deployed an **AI market-research platform** (live in production) that converts real-time market data into versioned, evidence-grounded trade theses via a **multi-agent debate pipeline** (analyst → bull/bear researchers → contradiction checker → trader → risk team → portfolio manager), streamed stage-by-stage to a React trading terminal.
- Built a **deterministic risk gate the LLM cannot override** (pure Python, fully unit-tested): reward:risk floors, confidence/coherence checks, stop-width guards, and latching daily-loss halts replicating prop-firm evaluation rules; every approval/rejection audit-logged with the specific failed rule.
- Architected a **risk-normalized evaluation framework** in R-multiples (expectancy, profit factor, **SQN**, per-trade Sharpe, **MFE/MAE excursions**, capture ratio, max-drawdown-in-R) sliced by regime/direction/asset, plus a **pre-registered memory-vs-baseline A/B experiment** with paired same-snapshot decisions.
- Engineered **real-time multi-asset ingestion** (crypto WebSockets, equities, FX, and macro via Coinbase, Polygon, Twelve Data, Finnhub, FRED) with replay adapters enforcing **point-in-time, no-look-ahead** guarantees for historical runs; **cost-aware model routing** (expensive judge models only where judgment is needed) under hard per-user spend caps.
- Shipped the full stack solo: FastAPI modular monolith with event bus, Postgres + pgvector, Redis, Alembic, Docker, JWT auth with plan tiers, CI/CD, and a **public transparency page** scoring every AI call, including losses.

Telaeris Inc — **May 2023 – Present**
Applied AI / Software Engineer — *San Diego, CA*

- Built end-to-end **agentic AI workflows** (intake → retrieval → reasoning → tool execution → guardrails → evaluation → deployment) using LangGraph, LangChain, OpenAI/Claude/Gemini APIs, structured outputs, evaluation datasets, and **LangSmith tracing**.
- Designed **multi-agent systems** for deep research, grounded search, browser automation, voice interaction, and document intelligence, including routing, memory, tool calling, human-in-the-loop review, and failure handling.
- Improved **search and recommendation quality** with hybrid retrieval (Pinecone embeddings, query understanding, metadata generation), raising **Hit@10 from 0.74 to 0.83** and **nDCG@20 by 3%** in A/B testing.
- Deployed containerized AI microservices on **AWS/EKS** (FastAPI, Docker, Kubernetes, CI/CD) with model-serving optimization and cost/latency improvements.

Rupeek Fintech Pvt Ltd — **Aug 2021 – Aug 2022**
Machine Learning Engineer — *Bangalore, India*

- Built **forecasting and risk pipelines** for **700+ loan portfolios** on Databricks/PySpark, reducing operational risk by **5.5%**; deployed XGBoost models with **drift-based retraining** and **SHAP explainability** under model governance.
- Developed **anomaly detection** (Isolation Forest, Autoencoders) improving fraud and risk monitoring by **11%**.
- Optimized **computer vision models** (U-Net, YOLOv3, ArUco calibration, ONNX), cutting inference latency **13s → 4s**; automated OCR/NLP extraction from appraisal documents at **95% accuracy** (Tesseract, Azure CV).

PUBLICATIONS & RESEARCH

- **When Structure Is Hidden: Measuring and Reducing the Compositional Gap in LLMs** (in preparation, EMNLP 2026)
- **On the Feasibility of Vision-Language Models for Time-Series Classification** (HICSS 2026)
- **CONFLATOR: Linguistic Structure in Code-Mixed Language Modeling** (EMNLP 2023)
- **PESTO: Dynamic and Relative Positional Encoding for Code-Mixed Languages** (AAAI 2022)
- **Marine Algae Stalk Ca^{2+} Regeneration Using Mimetic Differences** (mathematical modeling, CSRC 2024)

EDUCATION

San Diego State University — **Aug 2022 – May 2024**
M.S. in Computational Data Science — *San Diego, CA*

Indian Institute of Information Technology, Sri City — **Aug 2018 – Jun 2022**
B.Tech in Computer Science and Engineering — *India*

SKILLS

Quant / Research	Risk-normalized evaluation (expectancy, profit factor, SQN, Sharpe, MFE/MAE, drawdown), point-in-time backtesting, time-series forecasting, anomaly detection, A/B testing, statistical modeling
AI / ML	LLMs, RAG, Agentic AI, Multi-Agent Systems, Tool/Function Calling, LangGraph, LangChain, Guardrails, LLM Evaluation, RLHF/SFT, NLP, Computer Vision, Multimodal AI, Transformers
ML Frameworks	PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, Spark, ONNX, vLLM, SGLang, MLflow
Languages & Backend	Python, SQL, TypeScript, C++, FastAPI, Node.js, REST APIs, PostgreSQL, Redis, Kafka, WebSockets
Retrieval / Data	Pinecone, FAISS, pgvector, Qdrant, Weaviate, Elasticsearch, MongoDB, Databricks
Cloud / Infra	AWS, Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, Airflow, Prometheus/Grafana, OpenTelemetry